R Lab 8. Multivariate Regression

We'll be predicting the home sales price based on various characteristics of the home.
For most of our analysis, we can use the same commands as in the Univariate Regression, but notice
that the interpretation may be different.

```
> A = read.csv(url("https://dr-baron.github.io/415-615/Data/HOME_SALES.csv"))
> names(A)
 [1] "ID"              "SALES_PRICE"     "FINISHED_AREA"   "BEDROOMS"
 [5] "BATHROOMS"       "GARAGE_SIZE"     "YEAR_BUILT"      "STYLE"
 [9] "LOT_SIZE"        "AIR_CONDITIONER" "POOL"            "QUALITY"
[13] "HIGHWAY"
> attach(A)
> reg = lm(SALES_PRICE ~ FINISHED_AREA + BEDROOMS + BATHROOMS + GARAGE_SIZE + YEAR_BUILT, data=A )
> summary(reg)

Coefficients:
               Estimate Std. Error t value Pr(>|t|)
(Intercept)   -2.962e+03  4.171e+02  -7.101 4.13e-12 ***
FINISHED_AREA  1.276e-01  7.166e-03  17.806  < 2e-16 ***
BEDROOMS      -1.255e+01  3.894e+00  -3.223  0.00135 **
BATHROOMS      1.042e+01  4.945e+00   2.107  0.03561 *
GARAGE_SIZE    2.724e+01  5.930e+00   4.593 5.49e-06 ***
YEAR_BUILT     1.480e+00  2.153e-01   6.872 1.83e-11 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 71.26 on 516 degrees of freedom
Multiple R-squared:  0.7356,    Adjusted R-squared:  0.7331
F-statistic: 287.1 on 5 and 516 DF,  p-value: < 2.2e-16
```

What? A negative coefficient for the Bedrooms? A house with more bedrooms is cheaper?
Answer: yes, as long as the area of the house remains constant.

```
> anova(reg)
Analysis of Variance Table

Response: SALES_PRICE
               Df  Sum Sq Mean Sq   F value    Pr(>F)
FINISHED_AREA   1 6655486 6655486 1310.6215 < 2.2e-16 ***
BEDROOMS        1   27613   27613    5.4376   0.02009 *
BATHROOMS       1  142710  142710   28.1030 1.708e-07 ***
GARAGE_SIZE     1  224987  224987   44.3053 7.197e-11 ***
YEAR_BUILT      1  239808  239808   47.2239 1.832e-11 ***
Residuals     516 2620307    5078
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

FINISHED_AREA alone explains 6655486. BEDROOMS explains an additional amount of 27613. Etc.
Confidence intervals for the slopes.

```
> confint(reg, level=0.90)
```

```
                         5 %          95 %
(Intercept)    -3649.4960341 -2274.7356492
FINISHED_AREA      0.1157872     0.1394038
BEDROOMS         -18.9655253    -6.1333354
BATHROOMS          2.2702046    18.5680623
GARAGE_SIZE       17.4654181    37.0078645
YEAR_BUILT         1.1248812     1.8345057
```

Confidence intervals for the slopes with Bonferroni adjustment (just 5 slopes; suppose we are not interested in the interval for the intercept).

```
> confint(reg, level = 1 - 0.10/5)
                         1 %          99 %
(Intercept)    -3935.5690391 -1988.6626442
FINISHED_AREA      0.1108729     0.1443181
BEDROOMS         -21.6357675    -3.4630932
BATHROOMS         -1.1212061    21.9594731
GARAGE_SIZE       13.3988430    41.0744396
YEAR_BUILT         0.9772158     1.9821710
```

Testing several slopes in one hypothesis.
H_0: $\beta_4 = 0$ and $\beta_5 = 0$ vs H_1: either $\beta_4 \neq 0$ or $\beta_5 \neq 0$
Consider a reduced model without these variables. Compare two models via a partial F-test.

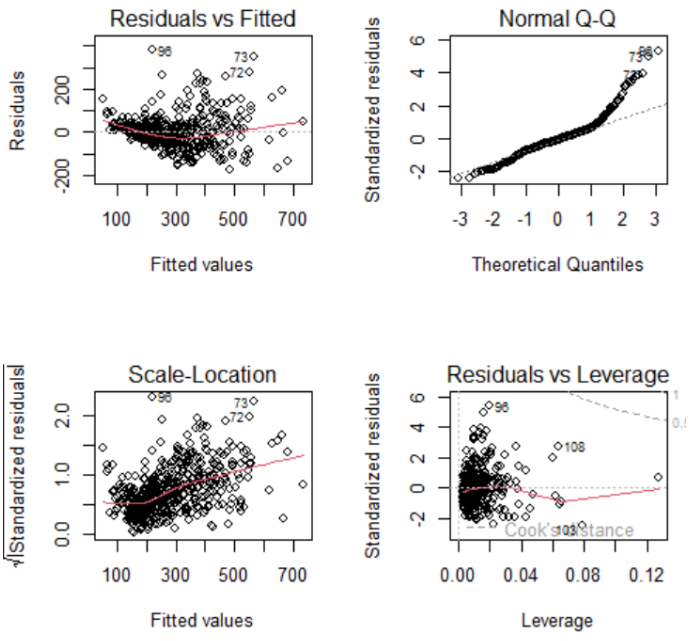
```
> reg.reduced = lm(SALES_PRICE ~ FINISHED_AREA + BEDROOMS + BATHROOMS, data=A )
> anova(reg.reduced, reg)
Analysis of Variance Table

Model 1: SALES_PRICE ~ FINISHED_AREA + BEDROOMS + BATHROOMS
Model 2: SALES_PRICE ~ FINISHED_AREA + BEDROOMS + BATHROOMS + GARAGE_SIZE + YEAR_BUILT

  Res.Df     RSS Df Sum of Sq      F    Pr(>F)
1    518 3085103
2    516 2620307  2    464796 45.765 < 2.2e-16 ***
```

Residual plots

```
> par(mfrow=c(2,2))
> plot(reg)
```



Testing a general linear hypothesis
H_0: $\beta_4 = 0$ and $\beta_5 = 0$

```
> C = rbind( c(0,0,0,0,1,0), c(0,0,0,0,0,1) )
> gamma = c(0,0)
> C
     [,1] [,2] [,3] [,4] [,5] [,6]
[1,]    0    0    0    0    1    0
[2,]    0    0    0    0    0    1

> library(car)
> linearHypothesis(reg, C, gamma)
Linear hypothesis test

Hypothesis:
GARAGE_SIZE = 0
YEAR_BUILT = 0
```

```
Model 1: restricted model
Model 2: SALES_PRICE ~ FINISHED_AREA + BEDROOMS + BATHROOMS + GARAGE_SIZE + YEAR_BUILT

  Res.Df     RSS Df Sum of Sq      F    Pr(>F)
1    518 3085103
2    516 2620307  2    464796 45.765 < 2.2e-16 ***
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1
```

> # Another example: test H_0: $\beta_4 = \beta_6 = \beta_7 = \beta_9$ in a model predicting mpg

```
>
> reg = lm( mpg ~ ., data=mtcars )
> coef(reg)
(Intercept)         cyl        disp          hp        drat          wt        qsec          vs          am        gear        carb
12.30337416 -0.11144048  0.01333524 -0.02148212  0.78711097 -3.71530393  0.82104075  0.31776281  2.52022689  0.65541302 -0.19941925
>
> C = rbind( c(0,0,0,0,1,0,-1,0,0,0,0),
+            c(0,0,0,0,0,0,1,-1,0,0,0),
+            c(0,0,0,0,0,0,0,1,0,-1,0) )
> C
     [,1] [,2] [,3] [,4] [,5] [,6] [,7] [,8] [,9] [,10] [,11]
[1,]    0    0    0    0    1    0   -1    0    0     0     0
[2,]    0    0    0    0    0    0    1   -1    0     0     0
[3,]    0    0    0    0    0    0    0    1    0    -1     0
>
> gamma = c(0,0,0)
>
> linearHypothesis(reg, C, gamma)
Linear hypothesis test

Hypothesis:
drat - qsec = 0
qsec - vs = 0
vs - gear = 0

Model 1: restricted model
Model 2: mpg ~ cyl + disp + hp + drat + wt + qsec + vs + am + gear + carb

  Res.Df    RSS Df Sum of Sq      F Pr(>F)
1     24 147.82
2     21 147.49  3   0.32269 0.0153 0.9973
```